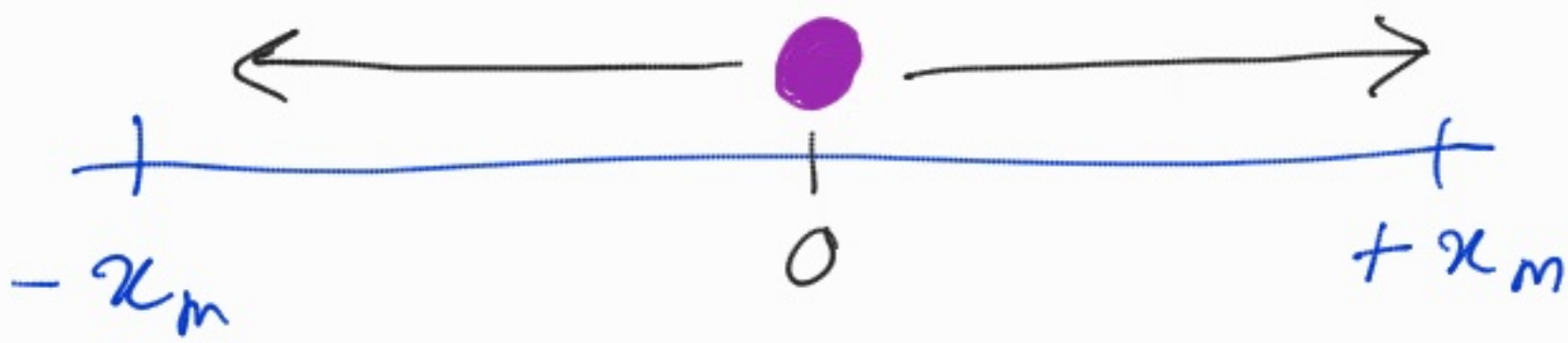
Simple Harmonic Motion

- Consider a particle oscillating from left to right along the x-axis:



- Frequency $\Rightarrow$ number of times per second it completes a full cycle.

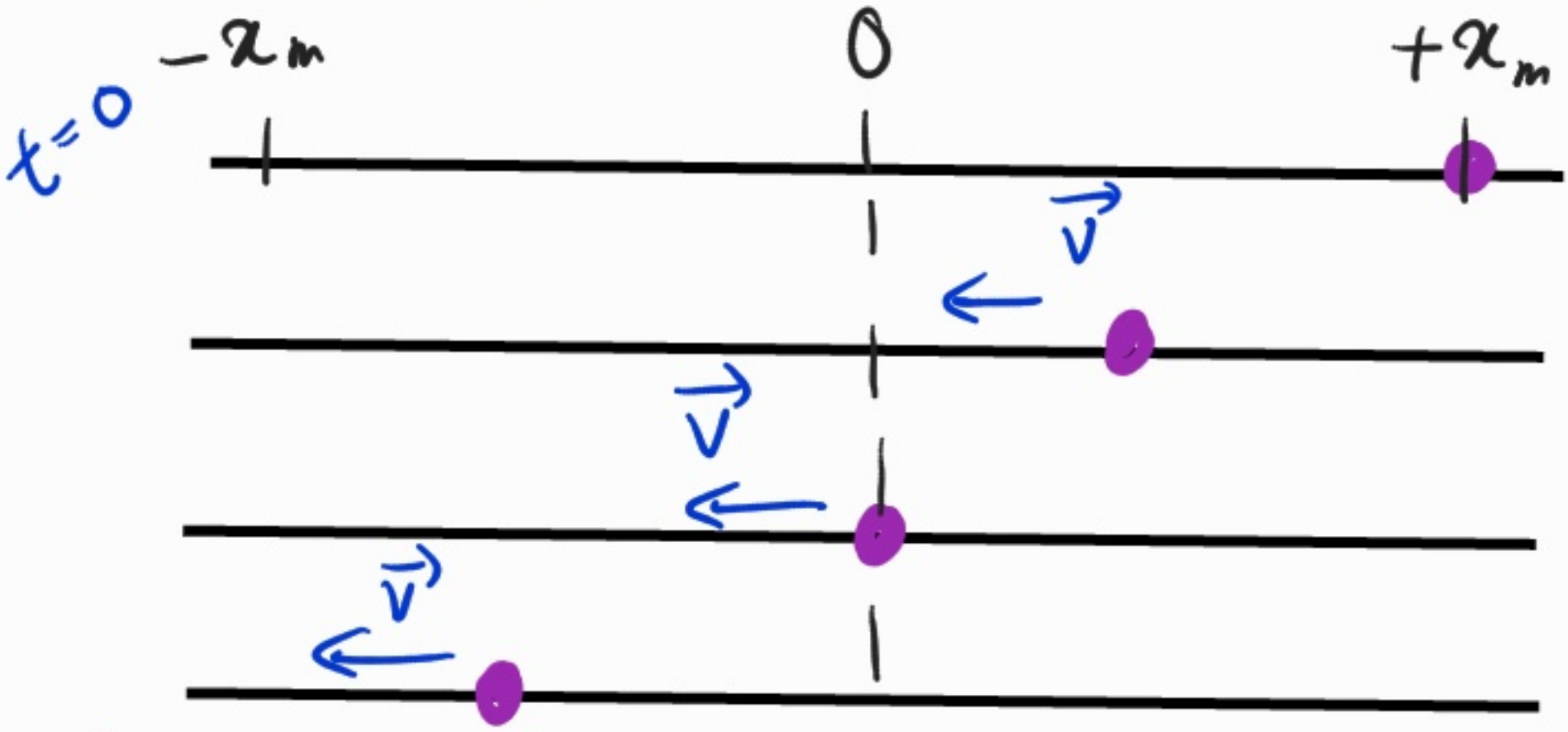
$$T = \frac{1}{f}$$

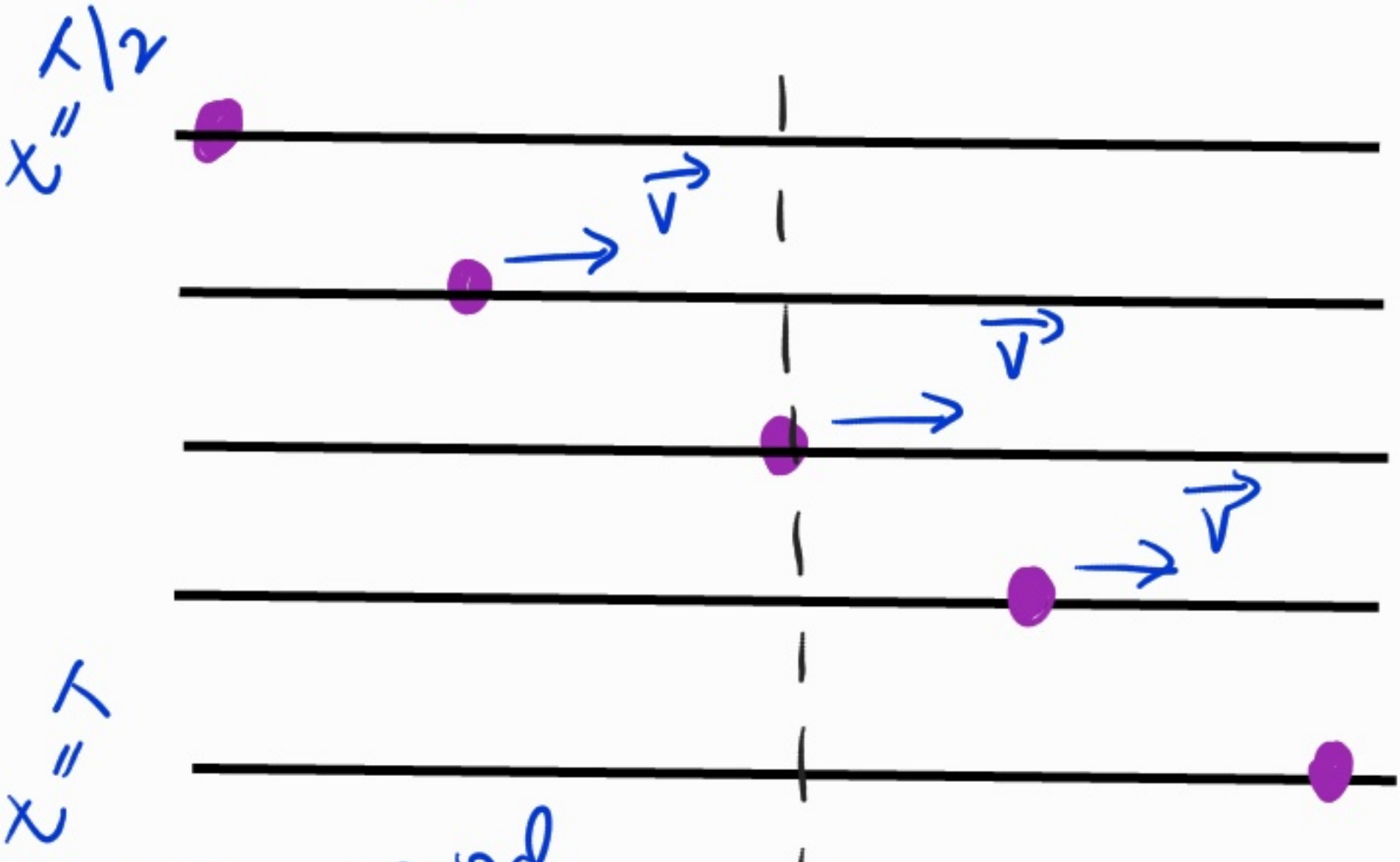
T → time for one full cycle

f → frequency

Simple Harmonic Motion

- Frame-by-frame Breakdown :



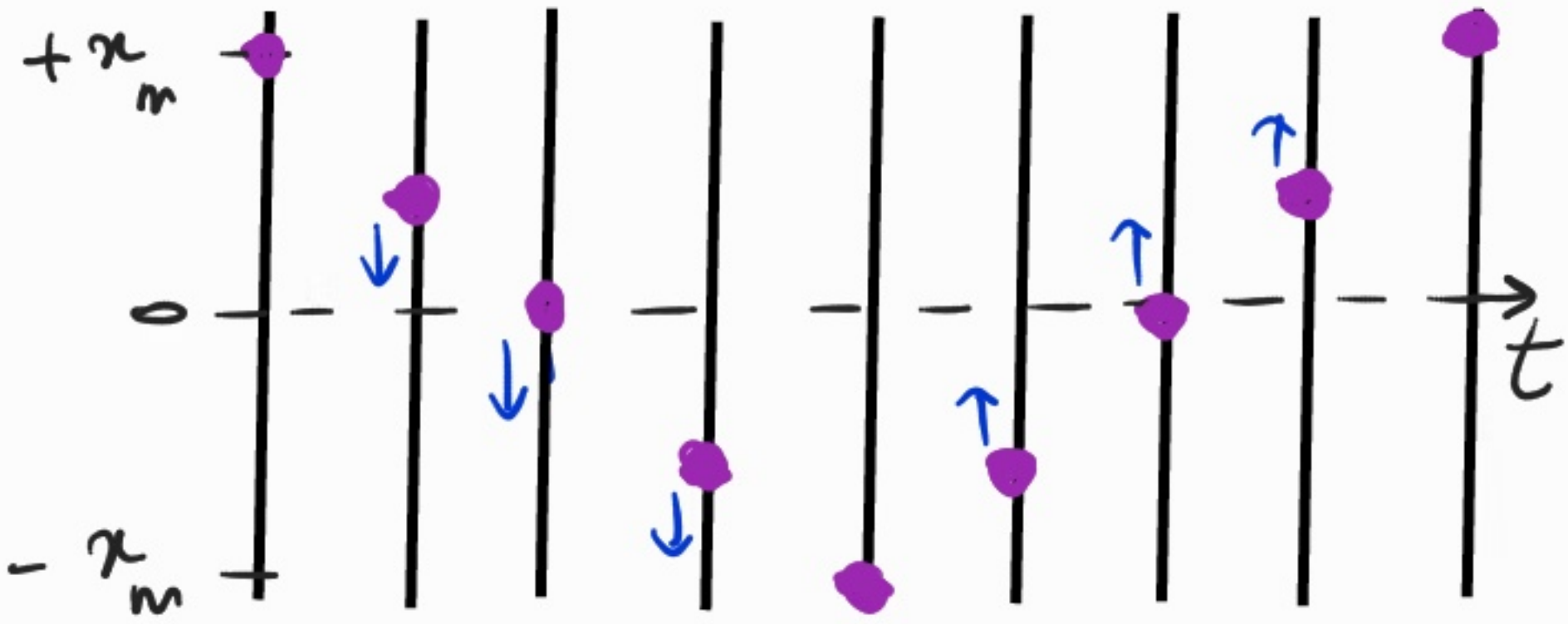


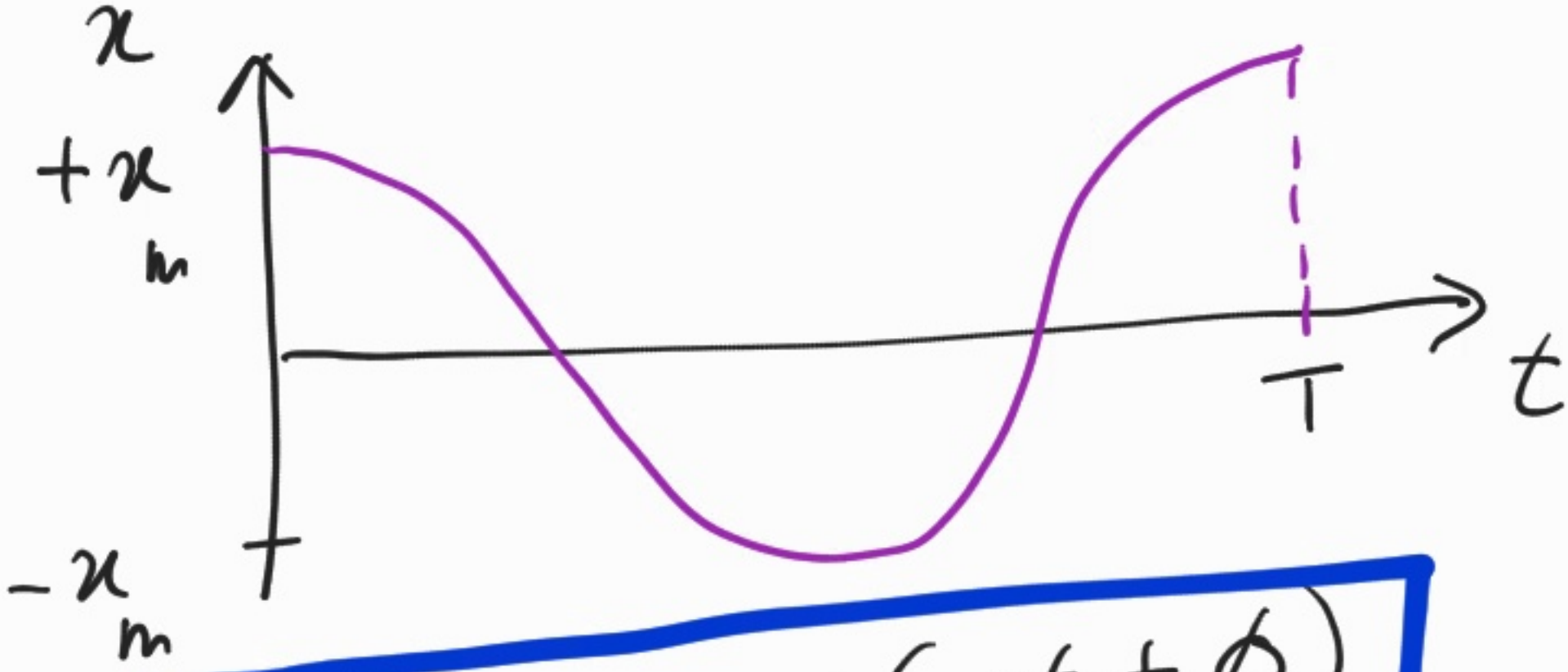
$T \sim$ period

$\vec{v} \sim$ particle velocity

Simple Harmonic Motion.

[ROTATED VIEW]





$$x(t) = x_m \cos(\omega t + \phi)$$

Mathematical Representation.

→ Let's look into this →

Simple Harmonic Motion

EQUATION OF MOTION

$$x(t) = x_m \cos(\omega t + \phi)$$

- x_m: amplitude
- ω: angular frequency
- t: time
- ϕ: phase constant

↳ particle displacement with respect to time

Angular frequency:

$\omega = \frac{\theta}{T}$ ⇒ for a full wave $\theta = 2\pi$ rad

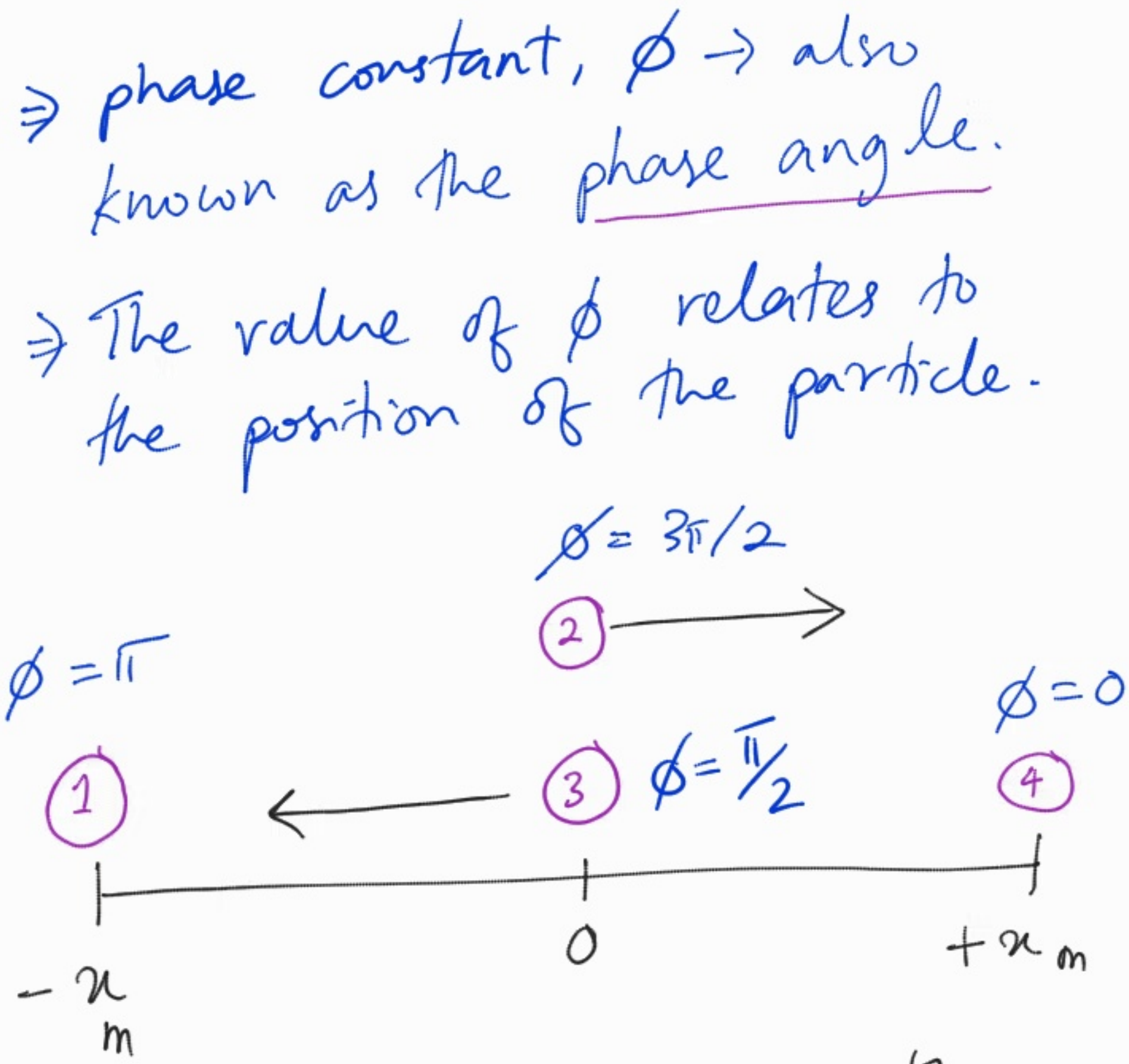
$$\therefore \quad \omega = \frac{2\pi}{T} = 2\pi f$$

Simple Harmonic Motion

- WHAT EXACTLY IS ϕ?

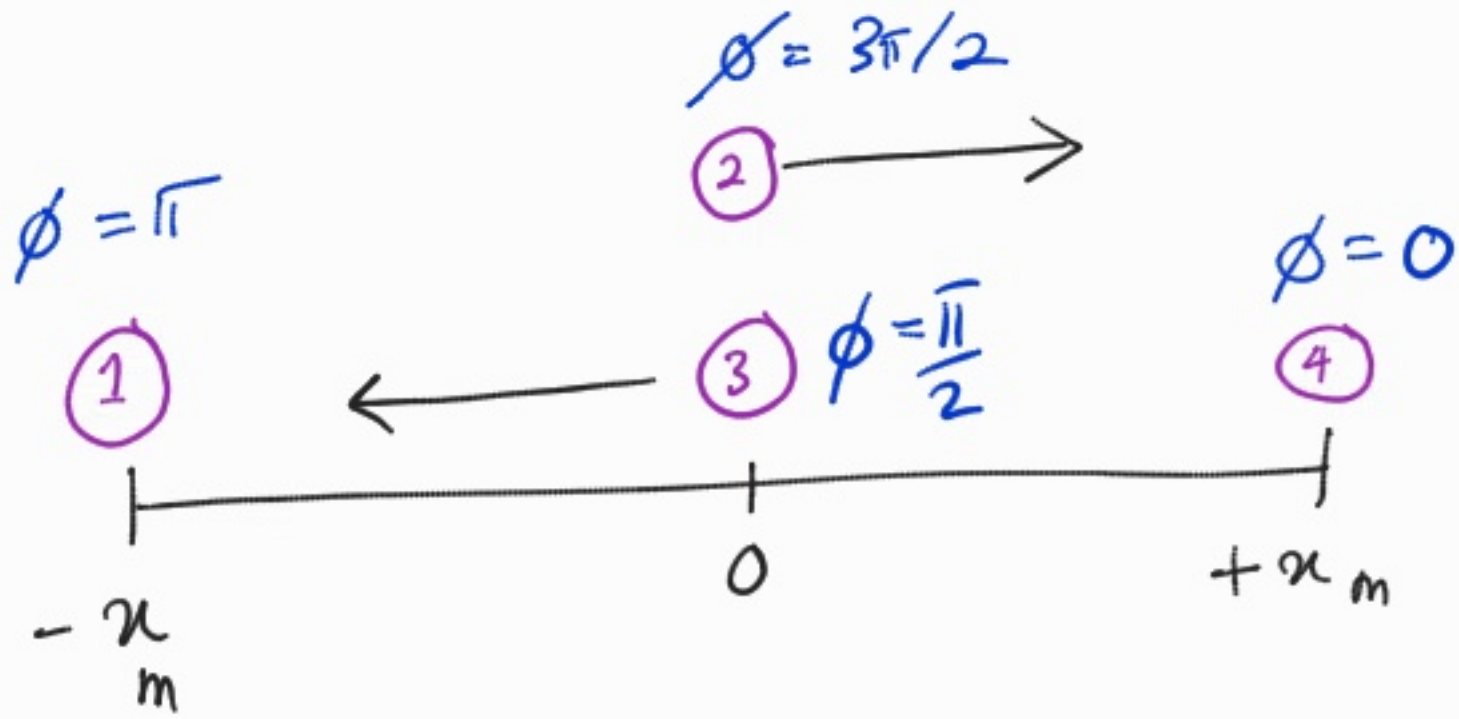
⇒ phase constant, ϕ → also known as the phase angle.

⇒ The value of ϕ relates to the position of the particle.



- Let's test this idea with our equation of motion at $t = 0$. (5)

Simple Harmonic Motion



$$x(t) = x_m \cos(\omega t + \phi)$$

1: $\phi = \pi$

$$x(0) = x_m \cos(0 + \phi) = x_m \cos(\pi)$$

$$= -x_m$$

3: $\phi = \pi/2$

$$x(0) = x_m \cdot \cos\left(0 + \frac{\pi}{2}\right) = 0$$

Try it for positions 2 & 4.

Simple Harmonic Motion

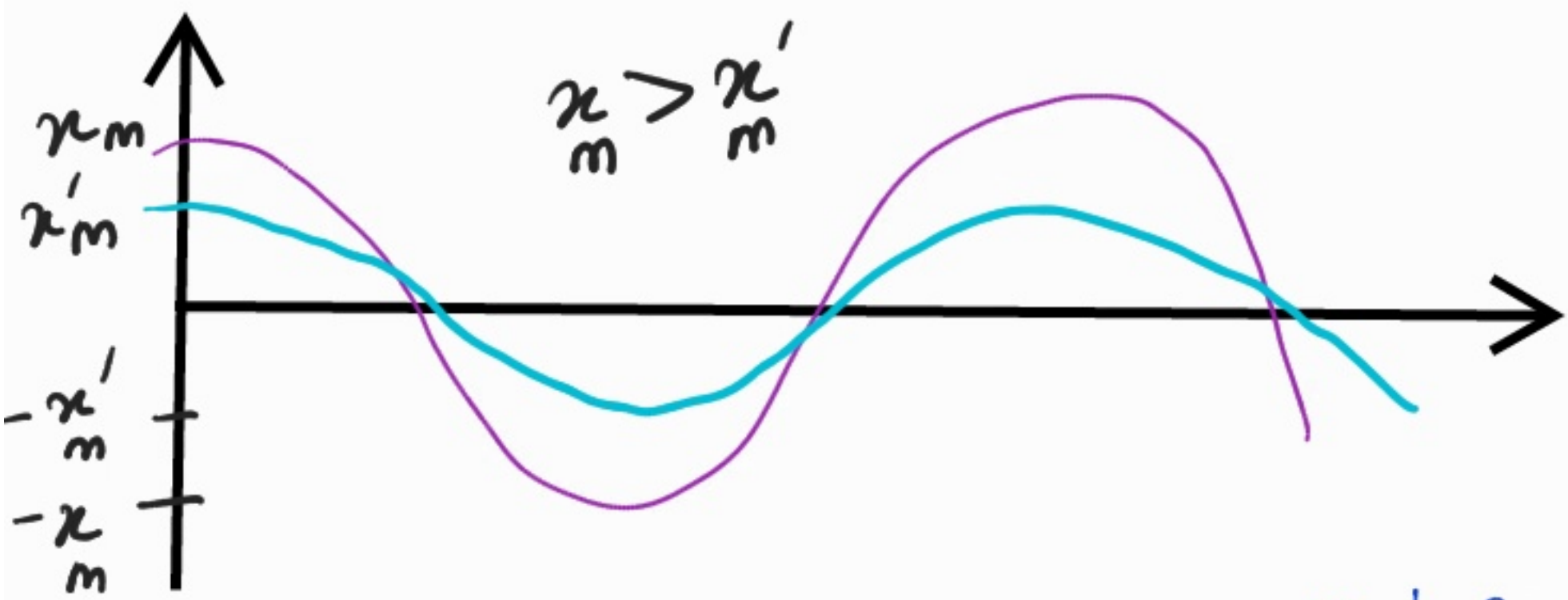
What does our analysis say?

- The initial position of the particle (at $t = 0$) is dependent on the phase angle, ϕ.

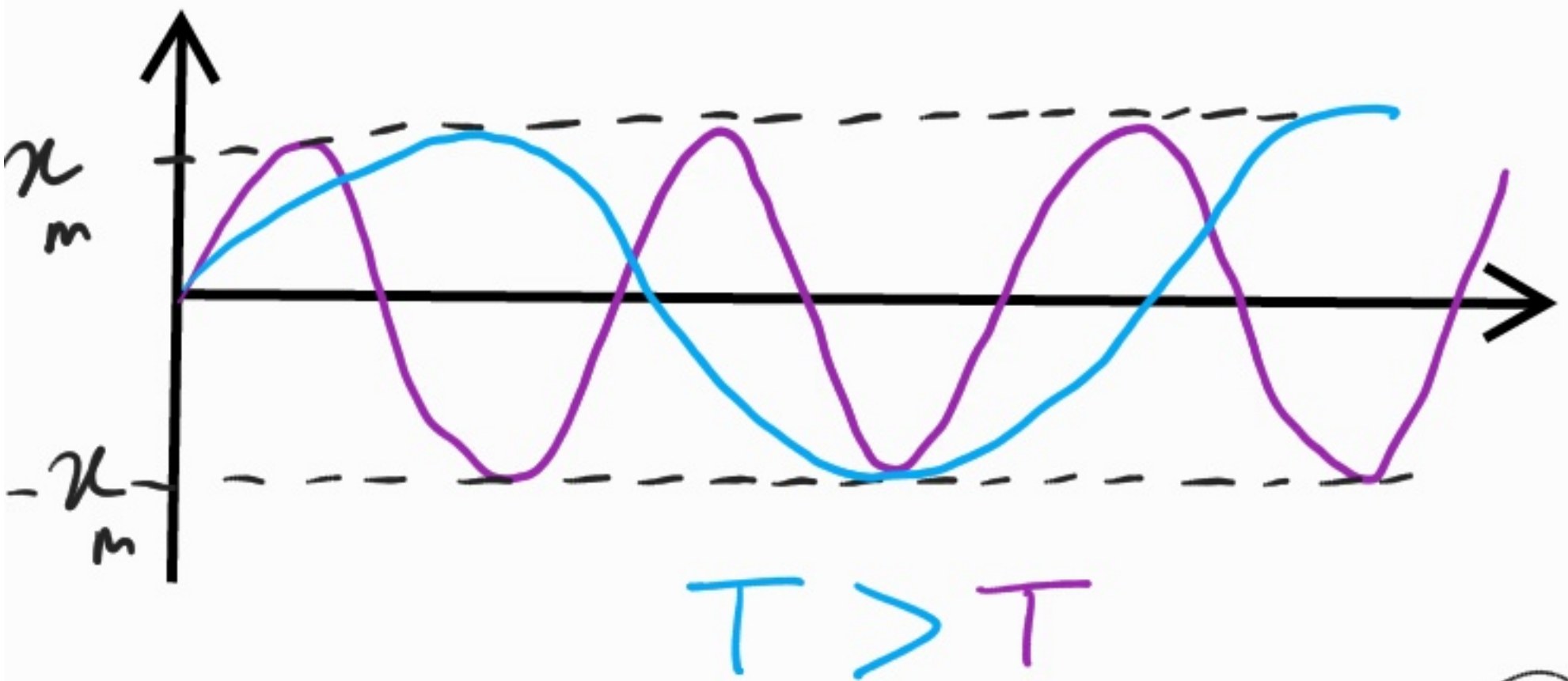
- At some specific time, t the phase angle, ϕ corresponds to the position of the particle.

Simple Harmonic Motion

- 2 oscillators : $A_1 \neq A_2$ but $f_1 = f_2$



- 2 oscillators : $A_1 = A_2$ but $f_1 \neq f_2$



Simple Harmonic Motion

→ Let's analyze the kinematics of the particle during its oscillation.

→ First let's get an expression for velocity:

$$x(t) = x_m \cdot \cos(\omega t + \phi)$$

$$v(t) = \frac{dx(t)}{dt} = \frac{d}{dt}\left[x_m \cdot \cos(\omega t + \phi) \right]$$

Let $Z = \cos(\omega t + \phi)$ — Apply Chain Rule

then $x = x_m Z$

$$\frac{dx}{dt} = \frac{dZ}{dt} \cdot \frac{dx}{dZ}$$

Let's find this expressions →

Simple Harmonic Motion

$$Z = \cos(\omega t + \phi) \quad \left\} \; x = \frac{x}{m} Z \right.$$

$$\frac{dZ}{dt} = -\omega \sin(\omega t + \phi)$$

$$\frac{dx}{dZ} = \frac{x}{m}$$

$$\therefore \; v(t) = \frac{dx(t)}{dt} = -\frac{x}{m}\omega \sin(\omega t + \phi)$$

→ Another way to represent this expression :

$$v(t) = -\frac{x}{m}\omega \cos\left(\omega t + \phi - \frac{\pi}{2}\right)$$

since $\cos\left(x - \frac{\pi}{2}\right) = \sin x$

Simple Harmonic Motion

→ Let's figure out the particle's acceleration:

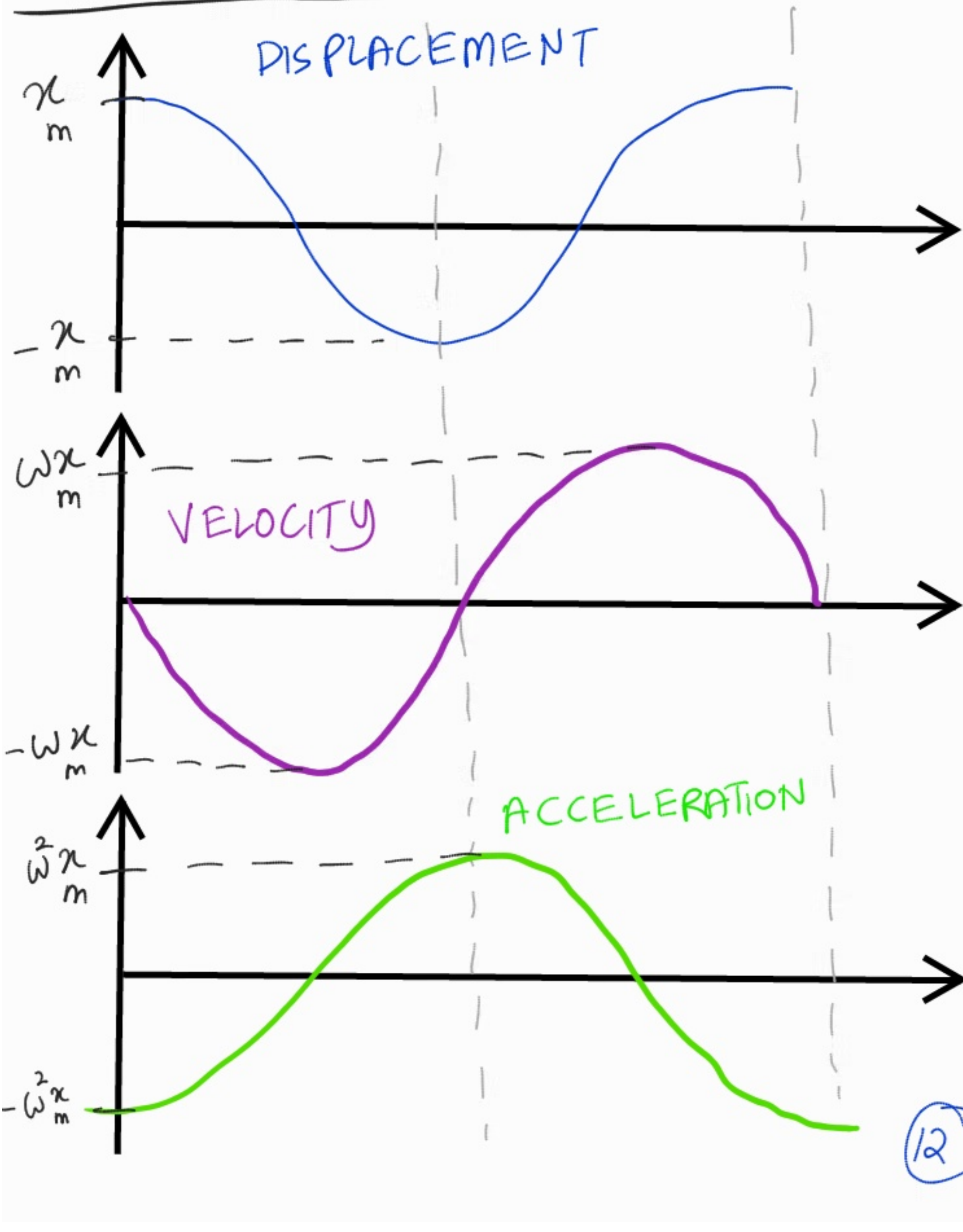
$$a(t) = \frac{dv(t)}{dt} = \frac{d}{dt}\left[-\frac{x}{m}\omega \cdot \sin(\omega t + \phi)\right]$$

$$= -\frac{x}{m}\omega \frac{d}{dt}\left[\sin(\omega t + \phi)\right]$$

$$= -\frac{x}{m}\omega^2 \cos(\omega t + \phi)$$

⇒ the amplitude for the acceleration is $\frac{x}{m}\omega^2$.

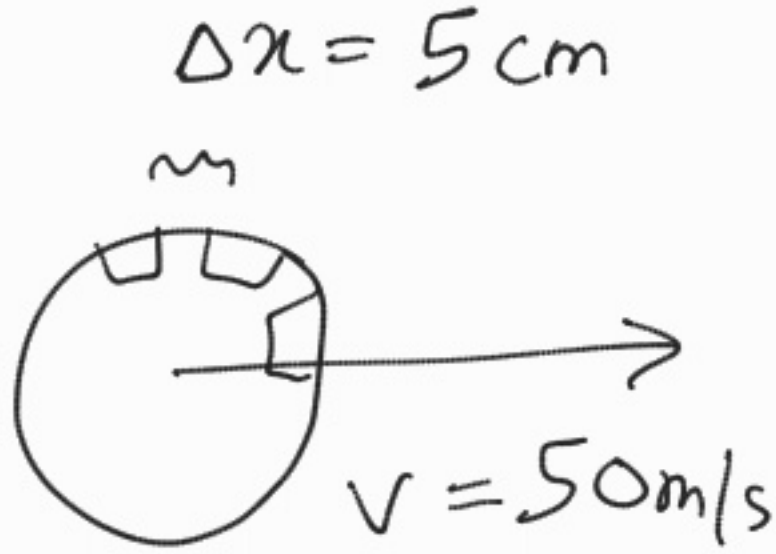
Simple Harmonic Motion



Practice Problem.

A car tire has a tread patterm where there is a 5 cm spacing between the crevices. As the tire moves, each crevice makes a single vibration. Determine the frequency of these vibrations if the car moves at the velocity of 50 m/s.

Solution:



$$v = \frac{\Delta x}{\Delta t}$$

frequency is : $f = \frac{1}{\Delta t}$

$$v = \Delta x f \rightarrow f = \frac{v}{\Delta x}$$

$$f = \frac{50\ m/s}{0.05\ m} = \underline{1\ kHz}$$